

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 11 · LECTURE NOTES

Depth, Matting, and Synthetic Defocus

Separating a scene into layers.

MODULE OUTCOMES

- Compare the methods of obtaining depth from a camera.
- Explain the matting equation and why it is underdetermined.
- Implement a matting method.
- Explain physically motivated synthetic defocus.
- Identify the failure modes and their causes.

1 Obtaining depth

Method	Mechanism	Characteristic failure
Stereo	Two cameras a known distance apart; disparity between corresponding points gives depth.	Textureless regions offer nothing to match, and occlusion boundaries have no correspondence because the surface is visible to only one camera.
Dual-pixel	Each photosite split into two halves, each seeing a different half of the aperture.	A baseline of millimetres , so depth resolution falls off rapidly beyond a metre or two. See below.
Depth from defocus	The size of the blur circle indicates distance from the focal plane.	Requires texture to measure blur against, and is ambiguous in sign — equally blurred in front of and behind focus — unless multiple focal settings are captured.
Active: time of flight, structured light	Project light and measure its return time or its deformation.	Defeated by sunlight , limited in range, and costs power and hardware. Excellent indoors at short range.
Learned monocular	A network predicts depth from a single image.	It is inference rather than measurement. See below.

Every one of these fails at the places that matter most for compositing — hair, fine structures, transparent objects, and occlusion boundaries. This is not coincidence: those are precisely the places where the assumption underlying each method (a locally continuous surface) breaks down. It leads directly into §2.

DUAL-PIXEL DEPTH IS WHY PORTRAIT MODE WORKS ON A SINGLE LENS

Each photosite is split into two independently readable halves, a design introduced for phase-detect autofocus rather than for depth.

The two halves see light arriving through slightly different parts of the lens aperture — which makes them a stereo pair with a baseline equal to roughly half the aperture diameter, a few millimetres.

So a single-lens camera produces a disparity map for free, with no additional hardware, no second camera, and no projected pattern. The sensor was already built that way.

The baseline is tiny, so depth resolution is poor beyond a metre or two. Which is entirely adequate, because portrait mode needs only to separate a subject at one or two metres from a background that is further away — it does not need metric depth. A good example of a method whose limitations happen not to matter for its application.

LEARNED MONOCULAR DEPTH IS INFERENCE, AND THAT IS WORTH SAYING

A single image does not determine depth. A small object nearby and a large object far away project identically, and no computation on the pixels distinguishes them. The problem is underdetermined in principle, not merely in practice.

So a monocular depth network is applying priors — learned regularities about how large objects usually are, how perspective lines converge, how texture density falls with distance, and what scene layouts are common.

It works well because those regularities are genuinely true of most photographs, and it fails when a scene violates them: a photograph of a photograph, a scale model, an unusual viewpoint, a mirror.

This is not a criticism of the method. Human monocular depth perception is also inference from priors, and it fails on the same cases — forced-perspective photographs fool people reliably. **The point is to be clear about which outputs of a pipeline are measurements and which are estimates**, which is Module 13's question and worth asking continuously.

2 Matting

$$\mathbf{I} = \alpha \mathbf{F} + (1 - \alpha) \mathbf{B}$$

Each observed pixel $\mathbf{I}$ is a blend of a foreground colour $\mathbf{F}$ and a background colour $\mathbf{B}$, in proportion α . **Three equations — one per colour channel — and seven unknowns per pixel: three for $\mathbf{F}$, three for $\mathbf{B}$, and α .** Underdetermined by four, at every pixel independently. **Every matting method is therefore a different way of supplying the missing constraints.**

A HARD MASK IS WRONG AT EVERY INTERESTING BOUNDARY

Hair, fur, motion blur, glass, smoke, and fine foliage are genuinely partially transparent at pixel scale. A pixel spanning a single hair strand really does contain light from both the hair and whatever is behind it — the mixture is physical, not an artefact of sampling.

A binary mask produces a hard cut, which reads immediately as a cut-out. It is the single most recognisable sign of a poor composite, and the eye detects it before the viewer can say why.

So α must be continuous, taking intermediate values wherever the pixel is genuinely mixed.

Recovering that continuous α is the matting problem, and it is substantially harder than segmentation — which only has to decide which side each pixel is on.

- **Trimap-based methods.** The user marks regions as definitely foreground, definitely background, and unknown. The method solves only within the unknown band, using the certain regions as constraints. This is the classical setting and it reduces the underdetermination substantially.
- **Bayesian matting.** Model the local distributions of foreground and background colours as Gaussians estimated from nearby known pixels, and solve for the most probable α , F , and B jointly.
- **Closed-form matting.** Assume F and B are each locally linear in colour within a small window — the 'colour line' model, which holds remarkably well for real images. **α then satisfies a sparse linear system whose matrix is a Laplacian**, which is the same structure as Module 08's Poisson system and CSCE 649's pressure solve. **A third appearance of the same machinery.**
- **Blue or green screen.** Constrain B to a known colour. The underdetermination is removed by controlling the scene rather than by inference, which is why film does it and why it works so much better than anything else.
- **Learned matting.** The best results now, requiring little or no user input — and inferring the matte from training data rather than deriving it.

3 Synthetic defocus

BLURRING BY DEPTH IS NOT WHAT A LENS DOES

A real lens integrates over its aperture. An out-of-focus point in the background is spread across a disc on the sensor — and crucially, **light from behind a foreground object reaches sensor positions beside that object**, because it arrives through parts of the aperture the foreground does not block. A blurred background wraps slightly *around* a sharp foreground.

Blurring the image by a depth-dependent radius does the opposite. A gather-based blur at a background pixel near the subject averages in foreground pixels, smearing the subject's colour outward into the background.

The symptom is a halo of foreground colour around the subject — a faint glow of skin tone or shirt colour in the blurred background. This is the classic signature of poor synthetic defocus and it is immediately recognisable once you know to look.

The correct formulation is scatter rather than gather: each source pixel contributes to a disc of destination pixels, with depth ordering respected so that nearer contributions occlude farther ones. More expensive, and physically right.

- **Scatter with depth ordering**, per the callout above.
- **Use a realistic aperture shape.** Real bokeh discs are polygonal, their shape set by the number of diaphragm blades, and they have a characteristic edge profile — bright-rimmed for some lens designs. A Gaussian blur looks nothing like it.
- **Operate in linear space** (Module 01). This is what produces the bright, distinct out-of-focus highlights people associate with fast lenses: a small very bright highlight spread over a disc remains bright in linear space and is washed out if the operation is done on encoded values. **Doing the blur in gamma space is the difference between convincing bokeh and grey smudges.**
- **Make the blur radius proportional to $|1/d - 1/d_{\text{focus}}|$,** which is the thin-lens relation — *not* proportional to depth, and not proportional to depth difference. The non-linearity matters: the falloff in front of the focal plane differs from behind it.
- **And handle the matte** (§2). The subject boundary needs a continuous α , or the composite has a cut-out edge however good the blur is.

THE FAILURES ARE ALL AT BOUNDARIES

Hair and fur. Matting is hardest exactly where the detail is finest, and portrait mode's subject is usually a person with hair. This is the most common visible failure.

Glasses and transparency. The depth is genuinely ambiguous — the glass is at one depth and what is seen through it is at another — and the matte is partial. Spectacle lenses are frequently blurred or sharpened wrongly.

Objects at the same depth as the subject are blurred or kept inconsistently, because the depth estimate cannot separate them and the segmentation must guess.

Thin structures. A railing, a branch, or a microphone stand passing behind the subject gets sliced where the depth map's resolution cannot resolve it.

And the overall giveaway is a uniformly blurred background with a uniformly sharp subject — which no real lens produces, because a real lens's blur varies continuously with depth throughout the scene, including across the subject itself. A real portrait has the near shoulder slightly softer than the eyes.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Wadhwa et al. — Synthetic Depth-of-Field with a Single-Camera Mobile Phone (free)	The portrait mode paper. Dual-pixel depth, segmentation, and the defocus rendering of §3, from the team that shipped it.
Levin, Lischinski & Weiss — A Closed-Form Solution to Natural Image Matting (free)	The matting Laplacian of §2. Elegant, and a third appearance of the same linear system.
Chuang et al. — A Bayesian Approach to Digital Matting (free)	The Bayesian formulation, and a clear statement of the underdetermination.
Szeliski — Computer Vision, Chapter 12 (free)	Stereo and depth estimation, with the failure analysis of §1.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Compute depth from a stereo pair and identify every region where it fails. Classify each failure as textureless or occluded.
- 2 If your phone exposes dual-pixel data, extract it and compute disparity. Measure the usable depth range.

- 3 Run a learned monocular depth model on a photograph of a photograph and document the failure.
- 4 Construct a forced-perspective photograph that fools both a depth network and a human.
- 5 Implement closed-form matting and extract a matte for a subject with hair. Compare against a binary segmentation.
- 6 Composite the same subject with a hard mask and with a continuous matte, and compare at the hair boundary.
- 7 Implement gather-based depth blur and produce the foreground halo artefact deliberately.
- 8 Implement scatter-based defocus with depth ordering and show the halo removed.
- 9 Implement a polygonal aperture and blur in linear and in gamma space. Photograph point highlights to make the difference obvious.
- 10 Collect five portrait mode photographs from any phone and identify the failure mode in each.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five sources of depth and the characteristic failure of each.
- 2 How does dual-pixel depth work, and why is its small baseline acceptable?
- 3 Why is monocular depth inference rather than measurement, and when does it fail?
- 4 Write the matting equation and count the equations and unknowns.
- 5 Why is a hard mask wrong, and where specifically?
- 6 Name five matting approaches and say which removes the underdetermination by construction.
- 7 Why does blurring by depth produce a halo, and what is the correct formulation?
- 8 Give five requirements for convincing synthetic bokeh.
- 9 Name five portrait mode failure modes and the overall giveaway.